

FILTER DESIGN TEST CASE

Band-Pass Filter - Band 30 (2300 MHz) for 4G LTE Carrier Aggregation
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Band-Pass Filter filter targeting Band 30 (2300 MHz) for 4G LTE Carrier Aggregation. The filter is designed on Langasite substrate with a target center frequency of 1193.9 MHz and bandwidth of 162.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Band-Pass Filter
- Frequency Band: Band 30 (2300 MHz)
- Application: 4G LTE Carrier Aggregation
- Substrate: Langasite
- Package: Flip-Chip BGA
- Design Tool: Ansys HFSS 2024R1
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Band-Pass Filter design targeting Band 30 (2300 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1193.9 MHz
Bandwidth (3 dB):	162.1 MHz
Insertion Loss Target:	≤ 1.4 dB
Return Loss Target:	≥ 18.0 dB
Out-of-Band Rejection:	≥ 22.0 dB
Isolation (Duplexer):	≥ 32.0 dB
Q Factor:	11895
Temperature Coefficient:	-11.4 ppm/°C
Die Size:	0.6 x 1.5 mm
Wafer Size:	6-inch
Number of Resonators:	6
Electrode Material:	W

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Ansys HFSS 2024R1 and load the Band-Pass Filter template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 1193.9 MHz, BW 162.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Langasite).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3582 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.4 dB
- Return loss in passband shall be >= 18.0 dB
- Out-of-band rejection at +/- 243 MHz offset >= 22.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 1.7 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 89%

6. TEST RESULTS

Measurement Date: 2025-01-23
Devices Tested: 73
Insertion Loss (meas): 1.14 dB
Return Loss (meas): 20.0 dB
Rejection (meas): 25.2 dB
Isolation (meas): 36.5 dB
Q Factor (meas): 11895
Group Delay Variation: 4.1 ns
Power Handling: +36 dBm
Die Yield: 88.0%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Band-Pass Filter design demonstrated excellent performance for Band 30 (2300 MHz).
- b) Insertion loss met target specification.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. Karen Mitchell	Date:	2025-02-02

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